

Project Documentation

C64 Expansion Board Extension Cartridge Side

Project number: 227

Revision: 0

Date: 25.05.2026



C64 Expansion Port Extension Cartridge Side Rev. 0

Module Description

Introduction

The purpose of the C64 Expansion Port Extension is to place an Expansion Port connector at a desired location and connect the actual C64 Expansion Port via a ribbon cable. The maximum length of said ribbon cable is not yet determined it is $>25\text{cm}$ and $<100\text{ cm}$. This board is providing the cartridge connector and a reset switch (two possible positions).

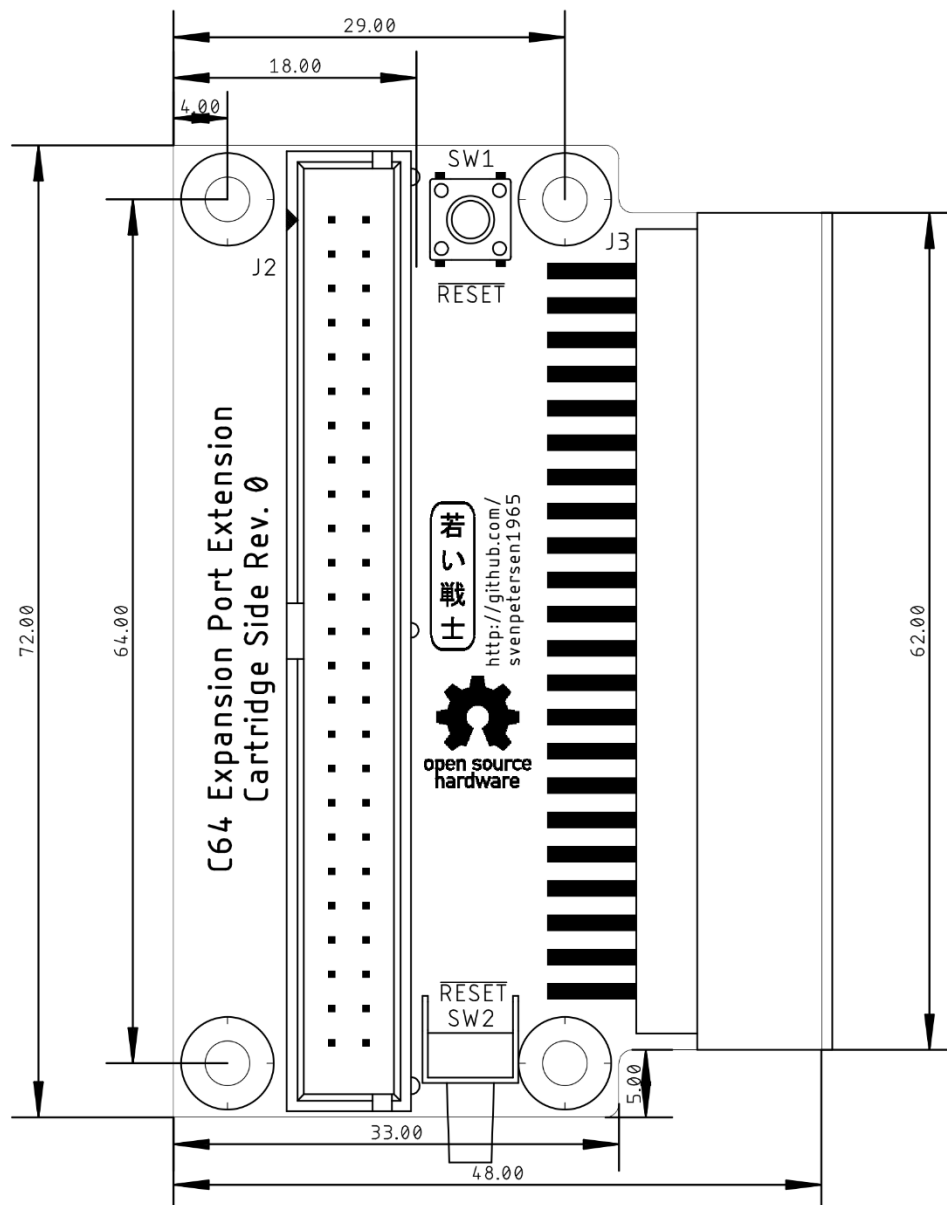


Figure 1: Dimension of the cartridge side PCB

Connector Pinouts

J2 – Break out connector

2x25 box header, 2.54mm pitch

Pin	Signal	Pin	Signal
1	GND	2	GND
3	+5V	4	GND
5	+5V	6	$\overline{\text{ROMH}}$
7	+5V	8	$\overline{\text{RESET}}$
9	$\overline{\text{IRQ}}$	10	$\overline{\text{NMI}}$
11	R / $\overline{\text{W}}$	12	PHI2
13	DOT CLOCK	14	A15
15	$\overline{\text{I/O1}}$	16	A14
17	$\overline{\text{GAME}}$	18	A13
19	$\overline{\text{EXROM}}$	20	A12
21	$\overline{\text{I/O2}}$	22	A11
23	$\overline{\text{ROML}}$	24	A10
25	BA	26	A9
27	$\overline{\text{DMA}}$	28	A8
29	D7	30	A7
31	D6	32	A6
33	D5	34	A5
35	D4	36	A4
37	D3	38	A3
39	D2	40	A2
41	D1	42	A1
43	D0	44	A0
45	GND	46	GND
47	GND	48	GND
49	GND	50	GND

J3 – Cartridge Connector

Card Edge Connector 22x2P 2.54mm (0.1") Right Angle

Pin (TOP)	Signal	Pin	Signal (BOT)
1	GND	A	GND
2	+5V	B	$\overline{\text{ROMH}}$
3	+5V	C	$\overline{\text{RESET}}$
4	$\overline{\text{IRQ}}$	D	$\overline{\text{NMI}}$
5	R / $\overline{\text{W}}$	E	PHI2
6	DOT CLOCK	F	A15

Pin (TOP)	Signal	Pin	Signal (BOT)
7	$\overline{\text{I/O1}}$	H	A14
8	$\overline{\text{GAME}}$	J	A13
9	$\overline{\text{EXROM}}$	K	A12
10	$\overline{\text{I/O2}}$	L	A11
11	$\overline{\text{ROML}}$	M	A10
12	BA	N	A9
13	$\overline{\text{DMA}}$	P	A8
14	D7	R	A7
15	D6	S	A6
16	D5	T	A5
17	D4	U	A4
18	D3	V	A3
19	D2	W	A2
20	D1	X	A1
21	D0	Y	A0
22	GND	Z	GND

Revision History

Rev. 0

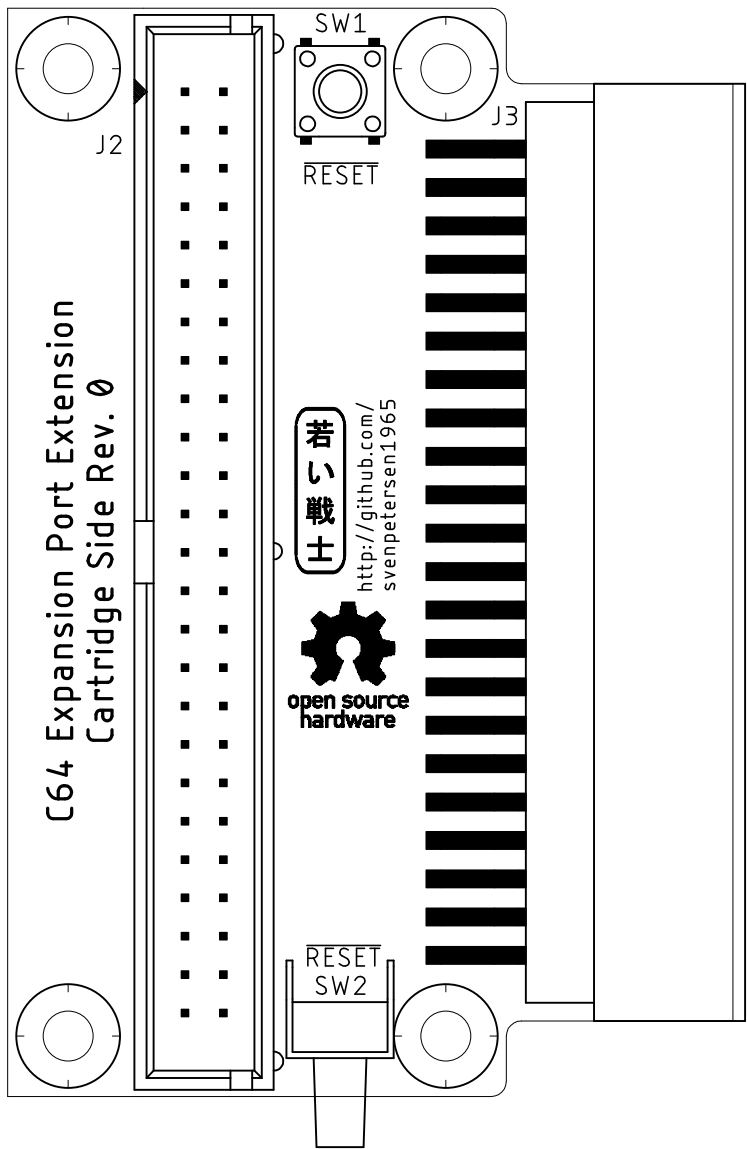
The prototype is functional with a 25cm ribbon cable.

See test description (Doc.-No. 227-6-02-00)

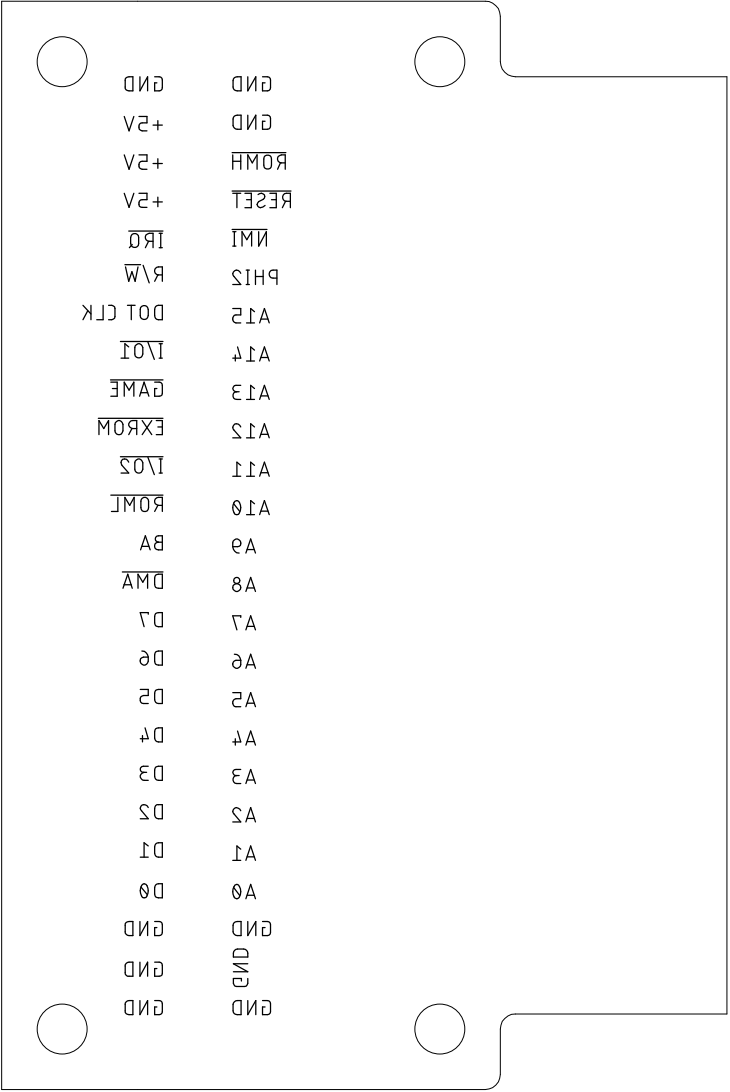


Title:	Break out for Expansion Port Cartridge Side	Doc.-No.:	227-1-01-00
Date:	22.05.2026 15:36	Draft.:	Sven Petersen
File:	C64_Exp_cartridge	Rev.:	0
	http://github.com/svenpetersen1965	Page	1/1
			A3

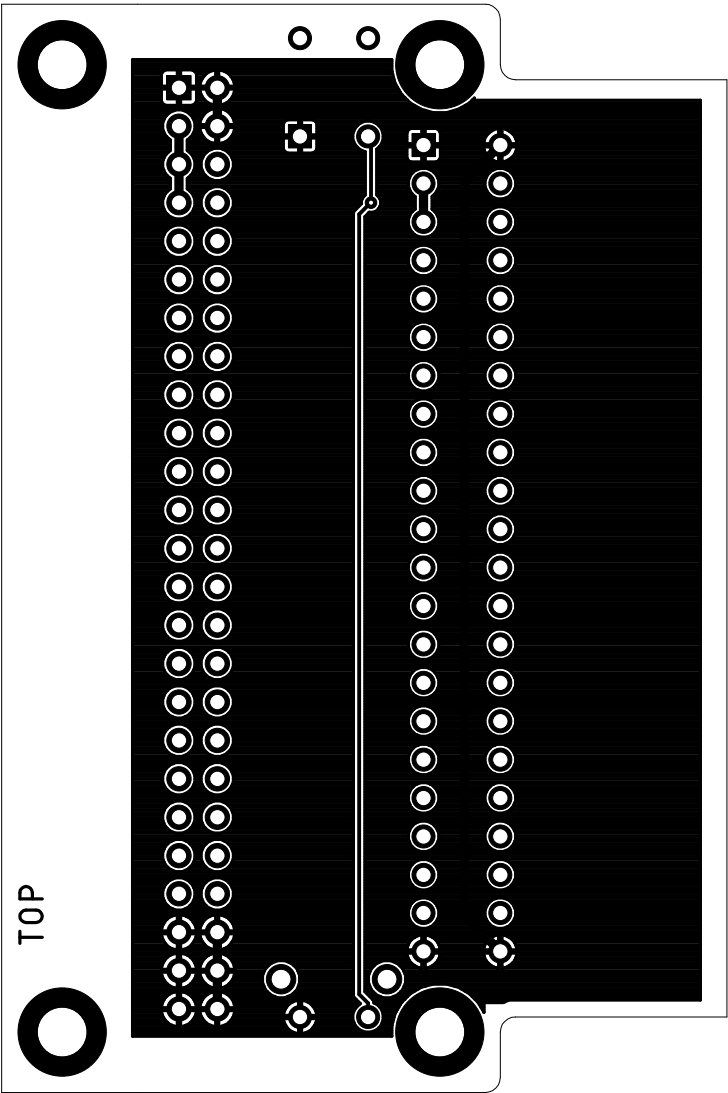
Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
placement component side		



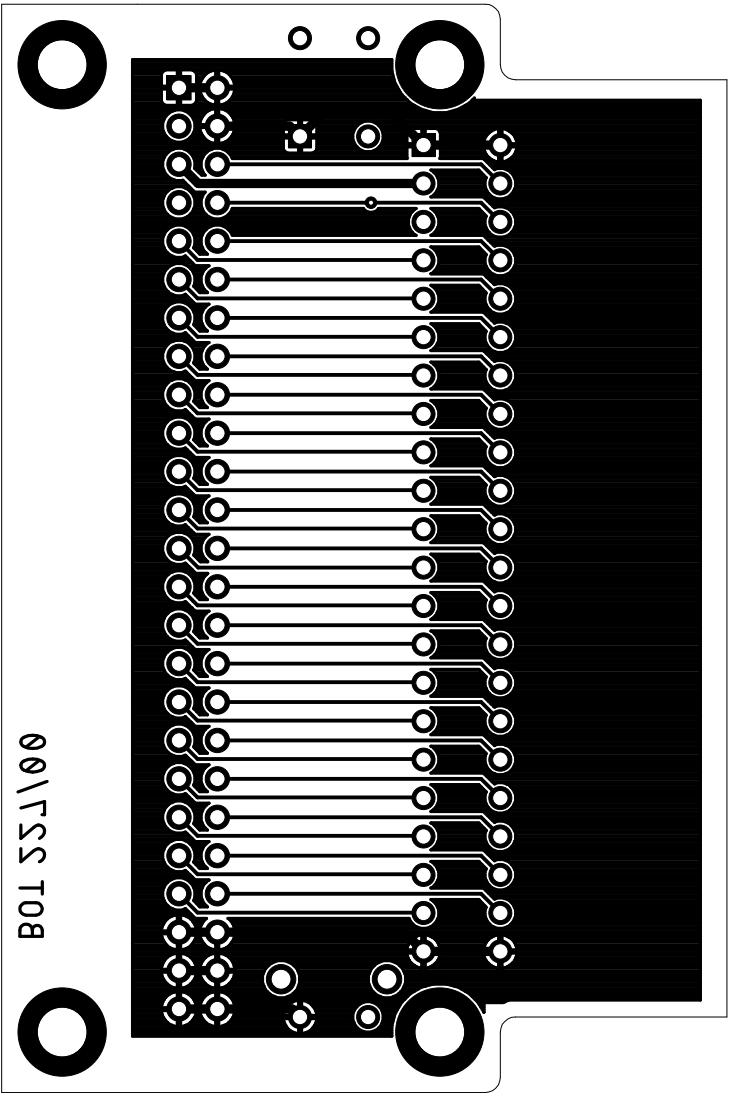
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C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
placement solder tabs		



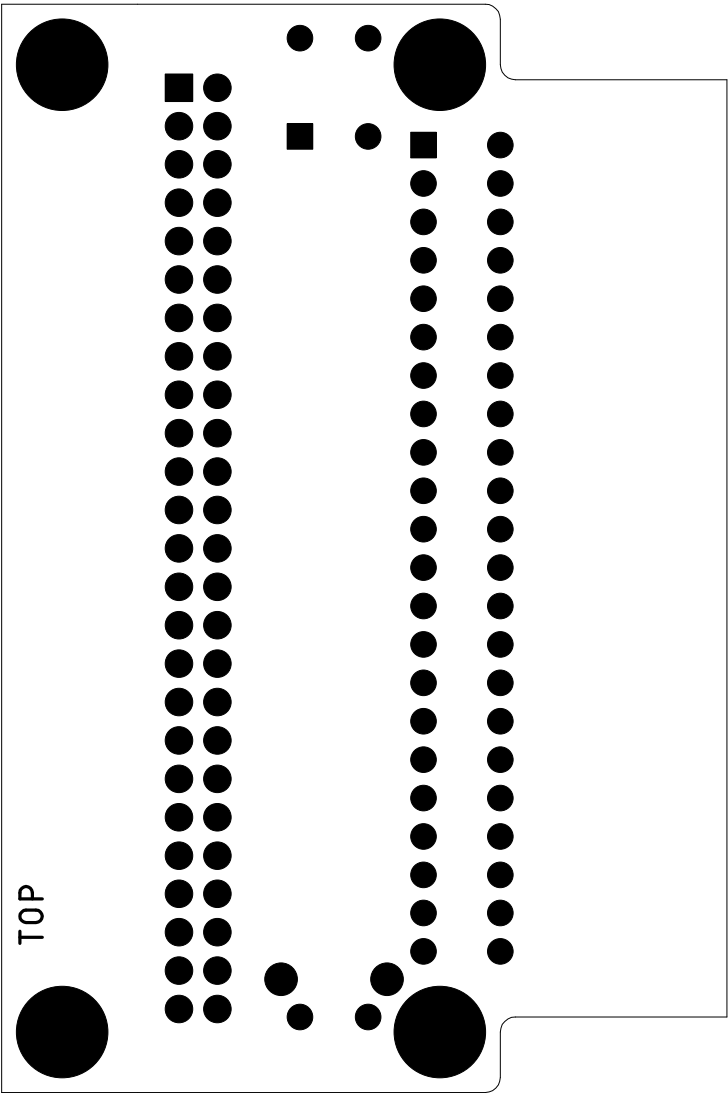
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22.05.2026 15:39		Rev.: 0
top		



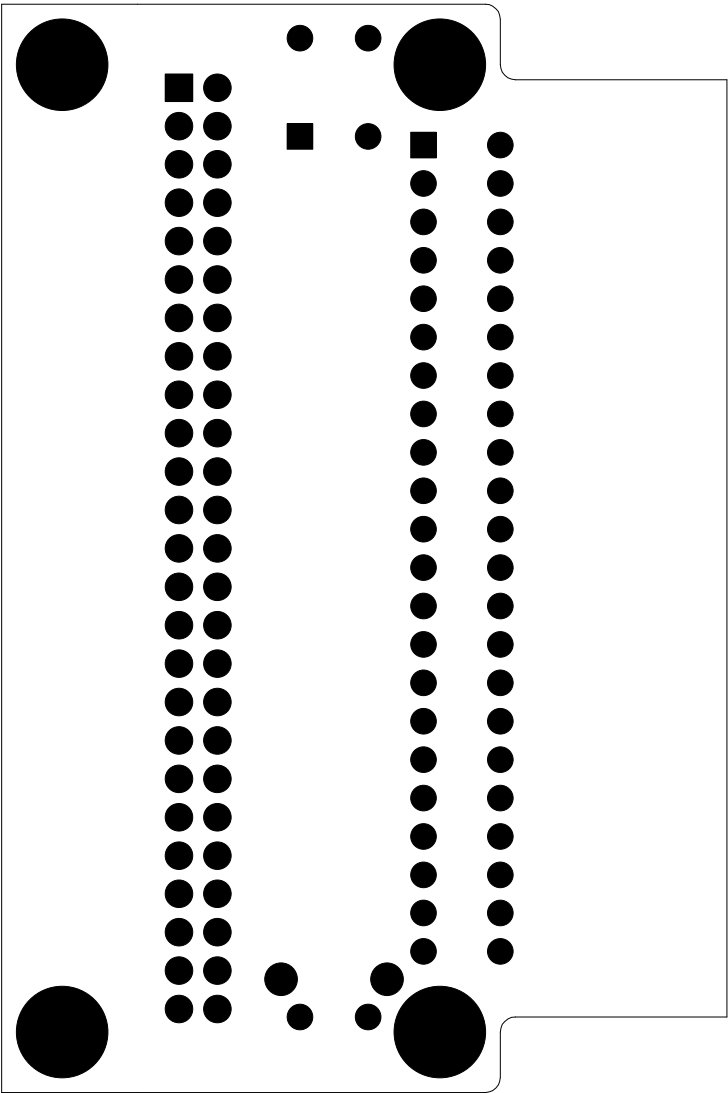
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bottom			



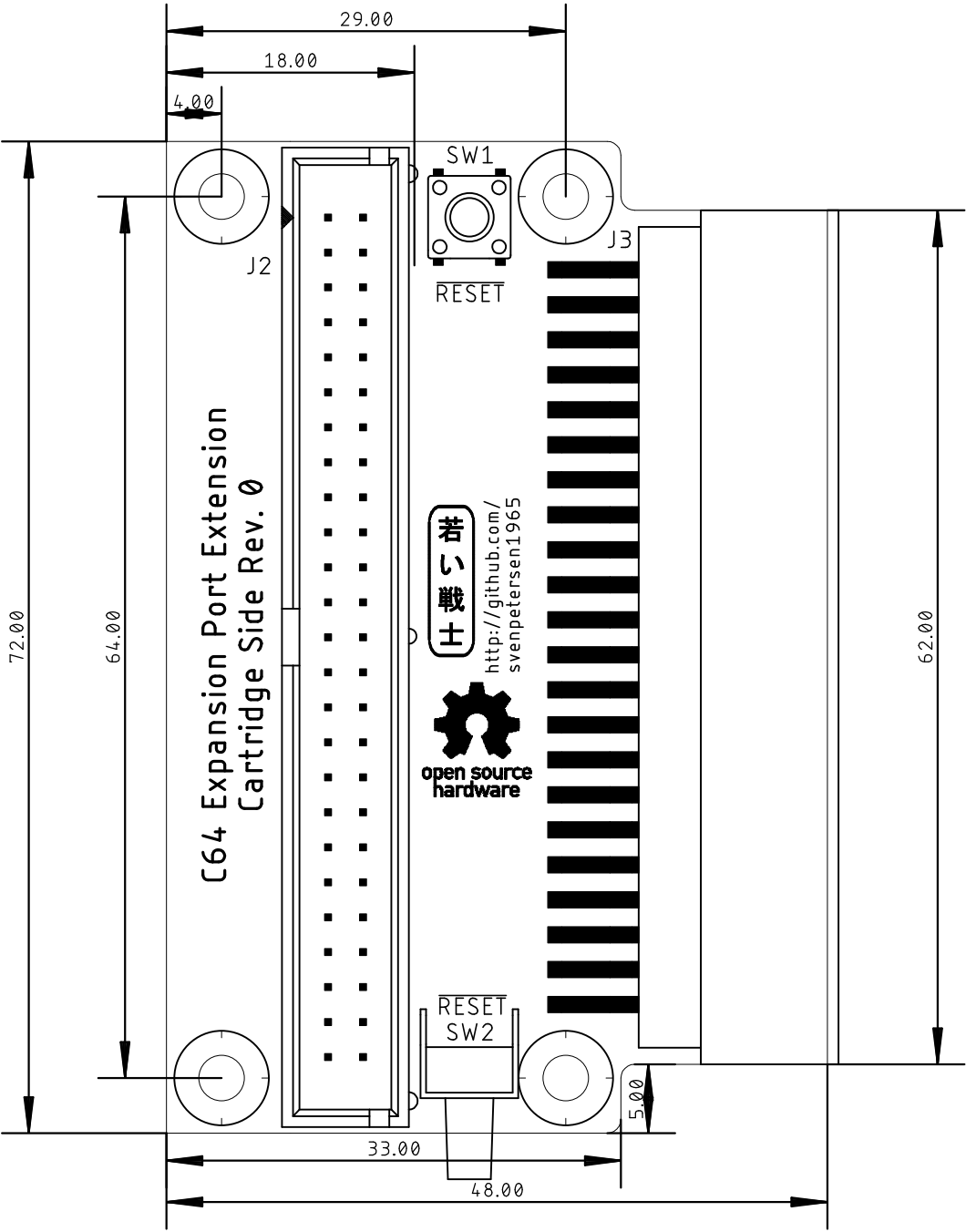
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22.05.2026 15:39		Rev.: 0
stopmask component side		

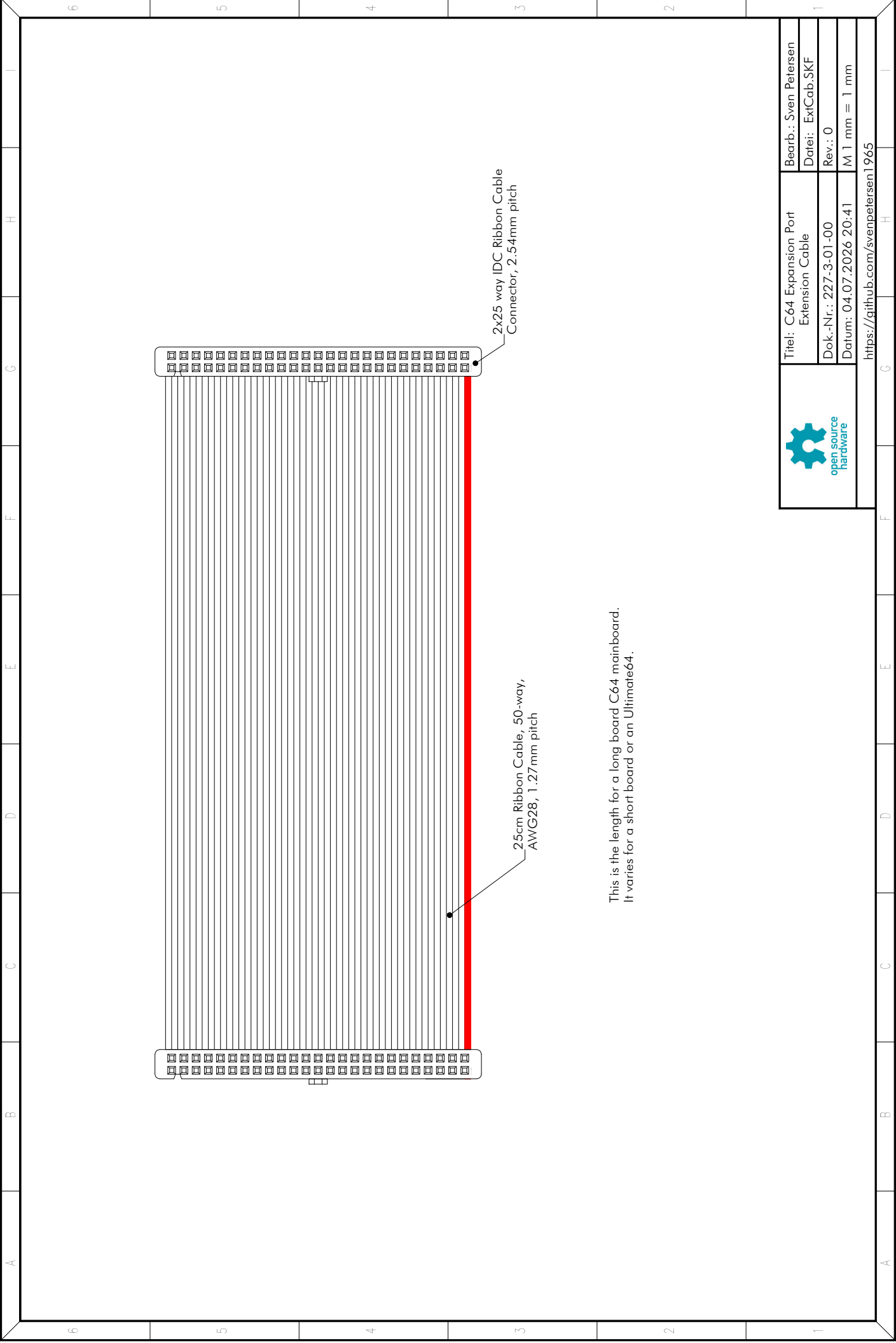


Sven Petersen 2026	Doc.-No.: 227-2-01-00	
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C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
stopmask solder side		



Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35μm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
placement component side		measures





	Titel: C64 Expansion Port Extension Cable		Bearb.: Sven Petersen
	Dok.-Nr.: 227-3-01-00		Datei: ExtCab.SKF
	Datum: 04.07.2026 20:41		Rev.: 0
	https://github.com/svenpetersen1965		M 1 mm = 1 mm

C64 Expansion Port Extension Cartridge Side Rev. 0

Test Description

Test Setup

- C64 short board
- C64 Expansion Port Extension, Port Side & Cartridge Side PCB Rev. 0
- Two 50-way ribbon cables, the lengths are 25cm and 100cm
- A variety of cartridges

Test *Execution*

The complete Expansion Port Extension was connected to the C64 expansion port on one side and the cartridge on the other side. Then, the C64 was switched on and different functions of the cartridge were tested.

Cartridge	25cm cable	100cm cable
Versa64cart	✓	✓
Easy Flash	✓	✓
Kung Fu Flash	✓	✓
Epyx Fastload	✓	✓
The Final Cartridge III	✓	✓
Ultimate II+	✓	✓
The Final Cartridge III+	✓	✗
Forth 64	✓	✗
Nordic Power 7.2	✓	✗
Nordic Power 2.1	✓	✗
Action Replay 6.0	✓	✗
Ms PAC-MAN	✓	✗
Power Cartridge	✓	✓

Conclusion

The Expansion Port Extension Rev. 0 is functional. The cable length is influencing the proper function of the cartridge, though. A 25cm ribbon cable is always working, a 100cm ribbon cable is not recommended. The maximum cable length ranges between those lengths.

C64 Expansion Port Extension Cartridge Side Rev. 0

Bill of Material Rev. 0.0

Pos.	Qty	Value	Footprint	Ref.-No.	Comment
1	1	227-2-01-00	2 Layer	PCB Rev. 0	2 layer, Cu 35μ, HASL, 72.0mm x 48.0mm, 1.6mm FR4
2	1	2x25 boxed header	2X25WV	J2	2x25, boxed pin header, e.g. reichelt.de WSL 40G
3	1	2x22 pin edge connector, 90°	EXPANSION_OU J3 T		1/5 x 5pc Industrial Card Edge Slot Socket Connector 22x2P 44P 2.54mm 0.1" Right Angle
4	1	JPT1236HB	JTP_1236HB	SW2	90° alternative reset switch, AliExpress search term "tact switch 90°"
5	1	JTP-1130	JTP-1130	SW1	Standard 6x6mm tact switch, e.g. Namae JTP-1130 or any other 6x6 tact switch, e.g. reichelt.de JTP-1130 or AliExpress search term "tact switch"
6	2	2x50 ribbon cable connector, 2.54mm			See drawing number 227-3-01-00, e.g. reichelt.de PFL 50
7	1	25cm 50 way AWG28, 1.27mm			ribbon cable